

Small Commercial Building Energy Load Analysis and Solar Feasibility Study

- 30-day load profile analysis with a simulated 30 kW rooftop solar PV scenario

Small Commercial Building Energy Analysis Dashboard

Made by Rohan Ramjutton

30-day load profile analysis with a 30 kW rooftop solar PV scenario

This dashboard shows how much electricity a small commercial building uses over a 30-day (monthly) period. The original building demand was compared with a simulated 30 kW rooftop solar system. In this project, the results obtained show that the solar generation can significantly reduce the amount of electricity imported from the grid during daytime hours. Most of the solar energy is found to have been directly used by the building, while excess solar is exported back to the grid when generation is higher than demand.

****Project assumptions have been listed at the bottom of this dashboard****

How to read these results:

- 1) kW measures power at a specific moment in time, similar to how fast electricity is being used.
- 2) kWh measures total energy used over time, this is what electricity bills are mainly based on.
- 3) Grid import means electricity brought in from the grid.
- 4) Solar self-consumed means solar energy used directly by the building.
- 5) Exported solar means extra solar energy sent back to the grid.

Key Metrics:	Value	Unit
Total Energy Used	8130.00	kWh
Peak Demand	34.90	kW
Estimated Bill before Solar	2731.20	\$
Estimated Emissions	5691.00	kg CO2
Overnight Base Load	2.39	kW

Solar Scenario	Value	Unit
Solar PV Size	30.00	kW
Grid Import After Solar	2548.65	kWh
Solar Self-Consumption	81.29	%
Monthly Solar Benefit	1626.99	\$
Simple Payback	1.84	years

Figure 1:

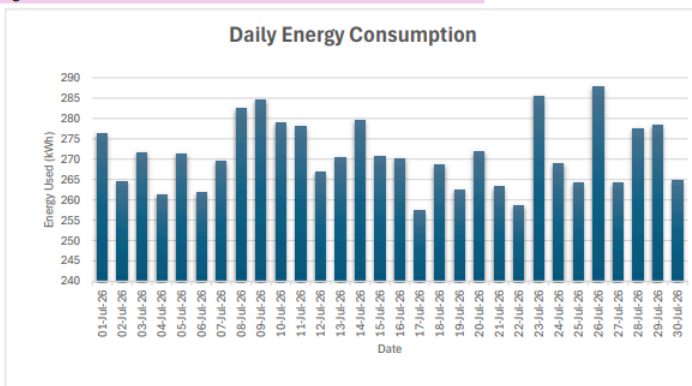


Figure 1 shows how much electricity the building used each day during the 30-day period. Higher bars indicate days with greater total energy use.

Figure 2:

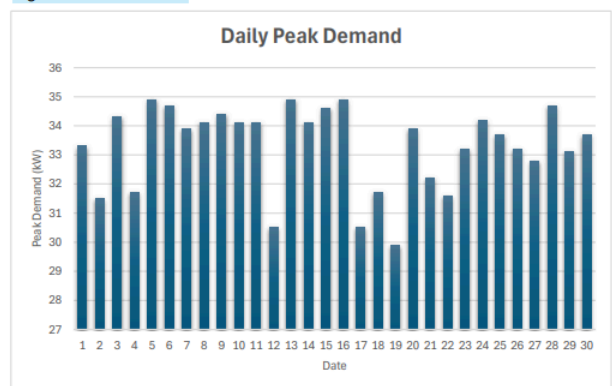


Figure 2 shows the higher power demand reached each day for 30 days. This is important as commercial sites may be charged based on peak demand (the amount of energy used during the peak hour)

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Brief summary of project:

This project analysed the electricity usage of a simulated small commercial building over a 30-day period using half-hourly load data. The aim was to understand the building's energy consumption profile, identify key demand patterns, estimate electricity cost and emissions, and assess the potential impact of a 30 kW rooftop solar PV system.

An Excel-based dashboard was developed to calculate total energy use, daily energy consumption, peak demand, load factor, overnight base load, estimated electricity cost and emissions. A simplified rooftop solar generation profile was then modelled and compared with the building demand to estimate solar self-consumption, exported energy, grid import after solar, monthly solar benefit and simple payback.

The analysis showed that the building used approximately 8130 kWh over the 30-day period, with a peak demand of 34.90 kW. Under the simulated 30 kW rooftop solar scenario, estimated grid import reduced to approximately 2549 kWh, with around 81.29% of generated solar energy used directly by the building. The estimated monthly solar benefit was approximately \$1627, giving a simple payback period of around 1.84 years under the assumed costs and tariffs.

This project demonstrates how electrical load data can be used to understand building energy behaviour and support early-stage solar feasibility assessment.

Project Aim & Scope:

Project aim:

The aim of this project was to develop an Excel-based energy analysis dashboard for a small commercial building and assess the potential impact of a rooftop solar PV system on grid electricity import and overall energy costs.

This project focused on:

- Analysing 30 days of half-hourly building demand data.
- Calculating daily energy consumption, peak demand and load factor.
- Estimating electricity costs and emissions.
- Identifying overnight base load.
- Modelling a simplified 30 kW rooftop solar PV generation profile.
- Estimating solar self-consumption, exported energy, grid import after solar and simple payback.

Project scope:

This was a concept-level energy analysis project. The load data used in this project was simulated to represent a small commercial building in a near-realistic scenario. The solar generation profile was also simplified and based on a typical daytime solar curve, with generation peaking around midday and falling to zero overnight.

The project does not include detailed electrical design, protection design, roof layout, weather variations, grid connection approval or any structural assessment. The purpose was to practice energy analysis, dashboard development and early-stage solar feasibility assessment.

Methodology:

Load Data:

The project used 30 days of half-hourly demand data. Each row represented one 30-minute interval.

$$\text{Energy (kWh)} = \text{Demand (kW)} * \text{Time (h)}$$

Since each interval was 30 minutes:

$$\text{Energy (kWh)} = \text{Demand (kW)} * 0.5$$

Daily summary calculations:

Excel formulas were used for the following:

Metric	Method
Daily energy	Sum of half-hourly kWh values for each day
Peak demand	Maximum kW value for each day
Average demand	Average kW value for each day
Minimum demand	Minimum kW value for each day
Load factor	Average demand divided by peak demand

Cost & Emissions:

Formulas used:

Energy cost = Total energy * Energy tariff

Demand cost = Maximum demand * demand charge

Supply charge = Number of days * daily supply charge

Estimated emissions = Total energy * emissions factor

Solar PV scenario:

Formulas used:

Net grid demand = Building demand - Solar generation

Solar self-consumed = MIN(building demand, Solar generation)

Exported solar = MAX(Solar generation - Building demand, 0)

Grid import = MAX(Building demand - Solar generation, 0)

Results:

The load analysis results were as follows:

Total energy used	8130 kWh
Peak demand	34.90 kW
Estimated bill before 30 kW solar	\$2731.20
Estimated emissions	5691 kg CO ₂
Overnight base load	2.39 kW

The building used approximately 8130 kWh over the 30-day period. The highest recorded demand was 34.90 kW, while the overnight base load was approximately 2.39 kW. The overnight base load suggests that some equipment remains active outside normal operating hours, such as standby equipment, refrigeration, security systems, lighting, servers or HVAC controls.

The solar PV scenario results were as follows:

Solar PV size	30 kW
Grid import after solar	2549 kWh
Solar self-consumption	81.29%
Monthly solar benefit	\$1626.99
Simple payback	1.84 years

The 30 kW rooftop solar PV scenario significantly reduced the building's grid electricity import. Before solar, the building required 8130 kWh from the grid over the 30-day period. After applying the simulated solar generation profile, grid import reduced to approximately 2549 kWh. Most of the solar energy was used directly by the building, with a self-consumption ratio of 81.29%. This indicates good alignment between solar generation and daytime building demand.

Discussion:

According to the results, the building demonstrates a daytime-heavy load profile, meaning that electricity demand generally increases during normal operating hours and reduces overnight. This type of load profile is suitable for rooftop solar PV because solar generation occurs mainly during the middle of the day, when the building is also using a higher amount of electricity. As a result, a large portion of the solar energy generated can be used directly by the building instead of being exported to the grid.

The solar analysis showed that the simulated 30 kW rooftop PV system reduced grid electricity import from approximately 8130 kWh to 2549 kWh over the 30-day period. This indicates that the PV system was able to offset a significant portion of the building's daytime energy demand. The self-consumption ratio of 81.29% also suggests strong alignment between solar generation and building usage, as most of the generated solar energy was consumed on site rather than exported. This is generally beneficial because using solar energy directly can provide greater value than exporting excess energy at a lower feed-in tariff.

The monthly energy comparison further highlights the impact of the solar scenario. Before solar, the building relied entirely on grid electricity to meet its demand. After applying the simulated solar generation profile, grid import was reduced substantially, showing the potential financial and energy benefits of rooftop PV for a commercial site with daytime operations. The estimated monthly solar benefit of approximately \$1627 and simple payback of 1.84 years suggest that, under the assumed tariff and system cost values, rooftop solar could be financially attractive for this type of building.

The analysis also identified an overnight base load of approximately 2.39 kW. Some overnight consumption is expected in a commercial building due to equipment such as refrigeration, security systems, standby appliances, lighting controls or HVAC systems. However, the presence of a consistent base load still represents a possible energy-saving opportunity. Reviewing after-hours equipment operation, standby loads and scheduling controls could help reduce unnecessary electricity use.

Overall, the results show that energy data can provide useful insight into how a building consumes electricity and where improvements may be possible. The dashboard helped identify daily energy use, peak demand, base load behaviour, estimated cost, emissions and the potential impact of solar PV. While the project uses simulated data and simplified assumptions, it demonstrates the basic process used in early-stage energy analysis and solar feasibility assessment. A more detailed version of this study could use real interval meter data, actual tariff information, weather-adjusted solar generation, inverter losses and site-specific electrical constraints.

Assumptions:

The following assumptions were used in this project:

- The load data was simulated for a small commercial building over a 30-day period.
- Electricity demand was recorded in 30-minute intervals.
- A 30 kW rooftop solar PV system was modelled using a simplified daytime generation curve.
- Solar generation was assumed to peak around midday and fall to zero overnight.
- Electricity tariff, demand charge, supply charge, feed-in tariff, emissions factor and solar system cost were assumed.
- The analysis does not include detailed electrical design, inverter losses, weather variation, shading, panel orientation, protection design, grid connection approval or roof structural assessment.

Conclusion:

This project developed an Excel-based energy analysis dashboard to assess the electricity demand profile of a simulated small commercial building over a 30-day period. The dashboard calculated key energy metrics including total energy consumption, peak demand, estimated electricity cost, emissions and overnight base load.

The results showed that the building used approximately 8130 kWh over the analysis period, with a peak demand of 34.90 kW. A simulated 30 kW rooftop solar PV system was then applied to assess the potential reduction in grid electricity import. The solar scenario reduced estimated grid import to approximately 2549 kWh, with 81.29% of generated solar energy used directly by the building.

Overall, the analysis showed that the building's daytime-heavy demand profile is well suited to rooftop solar PV. The project also highlighted the value of energy data in identifying demand patterns, estimating costs and supporting early-stage solar feasibility decisions. While the study used simulated data and simplified assumptions, it provided a practical introduction to commercial energy analysis, solar self-consumption and grid import reduction.